

SmartStudy: AI Learning Intervention System

Proposal for Shadow Enhancement

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Project: Shadow - Goal-Driven Academic Achievement System
Institution: Pan-Atlantic University
Date: November 2025
Purpose: Supervisor Approval for LLM Integration

Executive Summary

SmartStudy transforms Shadow from a passive academic tracking system into an active learning intervention platform using Large Language Models. When students struggle academically or want to learn better, SmartStudy provides personalized guidance, generates custom study plans, and measures whether interventions actually improve grades.

Core Innovation: Performance Detection → AI Diagnosis → Personalized Learning → Effectiveness Measurement

The Problem: Critical Gap in Academic Support Systems

Current Landscape Analysis

System	Strengths	Critical Limitation
Shadow (Current v1.0)	Tracks tasks, predicts CGPA, shows trajectory	✗ Tells students they're failing, not HOW to fix it
pauarchive	Centralized course materials	✗ No personalization, students don't know what to study
YouTube/Khan Academy	Free educational content	✗ Doesn't know individual student struggles
ChatGPT	Explains concepts on demand	✗ No academic context, no structured plans, no tracking
Traditional Tutoring	Personalized human guidance	✗ Limited availability, expensive, doesn't scale

The Core Student Problem

Students know **THAT** they're struggling, but not:

- WHY** - Root cause of confusion (concept gap? study method? time management?)
- WHAT** - Specific topics to focus on

- 3. **HOW** - Effective study approach for their learning style
- 4. **WHETHER** - If their efforts are actually working

Example Scenario:

- Student gets 45% on CSC401 Binary Trees test
- **Current Shadow:** "Warning: You're at risk of getting C in CSC401"
- **With SmartStudy:** "You struggled with recursion implementation. Here's a personalized 10-day recovery plan with videos, practice problems, and progress checkpoints. We'll track if this improves your next assessment."

SmartStudy Solution: Two-Mode AI System

Mode 1: Interactive AI Chat (Student-Initiated)

What It Is: An intelligent chat interface where students ask academic questions and receive context-aware responses.

How It's Different from ChatGPT:

Feature	Regular ChatGPT	SmartStudy Chat
Student context	None	Full academic history, courses, grades
Personalization	Generic	Tailored to learning style, performance patterns
Resource recommendations	None	Curated videos, practice problems, study materials
Effectiveness tracking	None	Monitors if explanations improved grades
Conversation persistence	Limited	Full history across sessions
Escalation capability	None	Suggests structured plans when needed

Example Interaction:

Student: "I don't understand recursion"

SmartStudy: "I see you recently scored 45% on your CSC401 Binary Trees assessment. Based on your test answers, you understand the concept but struggle with implementation, specifically base cases.

Since you're a visual learner, let me explain with animations...

[Personalized explanation]

Would you like me to:

1. Generate practice problems at your level

2. Create a 7-day mastery plan
3. Find the highest-rated video tutorials"

Key Innovation: Context-aware responses using student's actual academic data

Mode 2: Automated Intervention (AI-Initiated)

What It Is: Proactive system that detects struggles and generates intervention plans automatically.

How It Works:

1. **Detection:** Monitors performance for concerning patterns
2. **Diagnosis:** Analyzes why student is struggling (not just that they are)
3. **Prescription:** Generates personalized study plans with specific resources
4. **Tracking:** Measures whether intervention improved grades

Example Flow:

System detects: Student scored 45% on Binary Trees quiz
↓
AI analyzes: "Understands concept, struggles with implementation"
↓
Generates: 10-day recovery plan with daily tasks
↓
Delivers: Videos, practice problems, checkpoints
↓
Measures: Did next assessment score improve?
↓
Learns: "Visual approach worked best for this student"

Key Innovation: Closes the loop from detection to measurable improvement

Why Large Language Models Are Essential

1. Personalized Diagnosis

Without LLM: "You failed Binary Trees - study more"

With LLM: "You understand traversal logic but make off-by-one errors in implementation. Focus on edge cases and array indexing."

2. Custom Explanations

Without LLM: Generic tutorial link

With LLM: Explanation using PAU's specific terminology, student's exact confusion point, and their preferred

learning style

3. Adaptive Planning

Without LLM: Static "study for 2 weeks" plan

With LLM: Dynamic plan considering mood, available time, learning style, upcoming deadlines, and adjusts daily

4. Contextual Intelligence

Without LLM: Same recommendation for every student

With LLM: "You're a visual learner who studies best in mornings, you have 3 other deadlines this week - here's a 15min/day video-based plan"

Key Features & Shadow Improvements

Current Shadow Capabilities (v1.0)

- ✔ Task management with CA tracking
- ✔ CGPA calculation & prediction
- ✔ Course risk indicators
- ✔ Basic mood logging
- ✔ Rule-based priority recommendations

SmartStudy Enhancements (v2.0)

Feature	Description	Student Impact
AI Chat Interface	Ask questions, get context-aware answers	Immediate help when confused
Performance Diagnosis	AI analyzes WHY student struggles	Identifies root cause, not symptom
Personalized Study Plans	7-14 day learning paths with daily tasks	Clear roadmap to improvement
Intelligent Curation	Auto-discovers best YouTube videos, Reddit posts	Always finds highest quality resources
Adaptive Reminders	Mood-aware, context-aware notifications	Help when needed, not spam
Effectiveness Tracking	Measures grade improvement from interventions	Proves what works, adapts approach
Multi-Format Delivery	Videos, text, practice problems, AI explanations	Matches learning style automatically
Cross-Course Learning	Identifies knowledge gaps across courses	Connects related concepts

SmartStudy Access Points (Not Just for Failing Students)

SmartStudy serves **all students in four scenarios:**

1. Preventive (Before Assessment)

Trigger: "I have a test next week"

Response: 7-day comprehensive preparation plan

2. Proactive (Want Deeper Understanding)

Trigger: "I passed but want to learn this better"

Response: Enrichment plan with advanced concepts

3. Reactive (After Struggling)

Trigger: Student scored below target

Response: Intervention plan to catch up

4. Exploratory (Curiosity-Driven)

Trigger: "I want to learn about X"

Response: Self-directed learning path

Key Point: SmartStudy is for all students, not just struggling ones.

Implementation Timeline (6 Months)

Phase 1: Interactive Chat (Month 1 - December 2025)

Focus: Build AI chat interface with academic context

Deliverables:

- Chat interface in Shadow dashboard
- GPT-4 integration for context-aware responses
- Conversation history and persistence
- Basic resource recommendations

Milestone: Students can ask SmartStudy questions and receive personalized academic guidance

Phase 2: Enhanced Intelligence (Month 2 - January 2026)

Focus: Add automation and content curation

Deliverables:

- Automatic struggle detection
- Study plan generation system
- YouTube/Reddit content curation algorithm
- pauarchive integration (when available)

Milestone: Automated intervention working end-to-end

Phase 3: Learning Loop (Month 3 - February 2026)

Focus: Tracking and adaptation

Deliverables:

- Smart notification system
- Effectiveness tracking dashboard
- Adaptive learning algorithms
- System polish and integration

Milestone: Complete learning loop with measurable outcomes

Phase 4-6: Beta Testing (March-June 2026)

Focus: Real-world validation

Activities:

- Deploy to 20-30 PAU students
- Collect effectiveness data
- Refine based on real usage patterns
- Document grade improvements
- Gather user testimonials

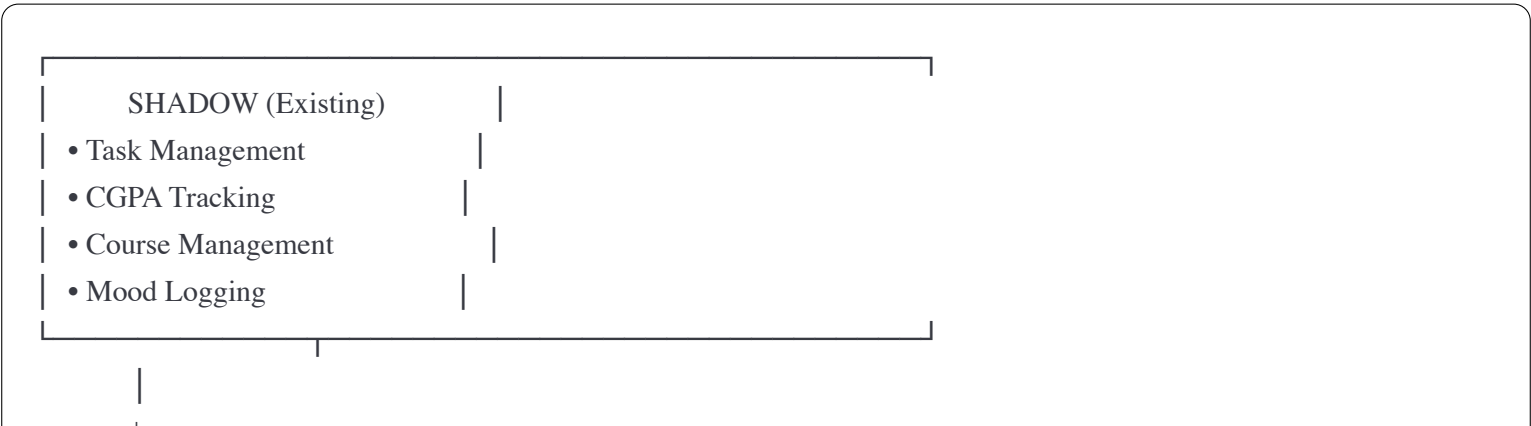
Milestone: Defense-ready with 3-4 months of real user data

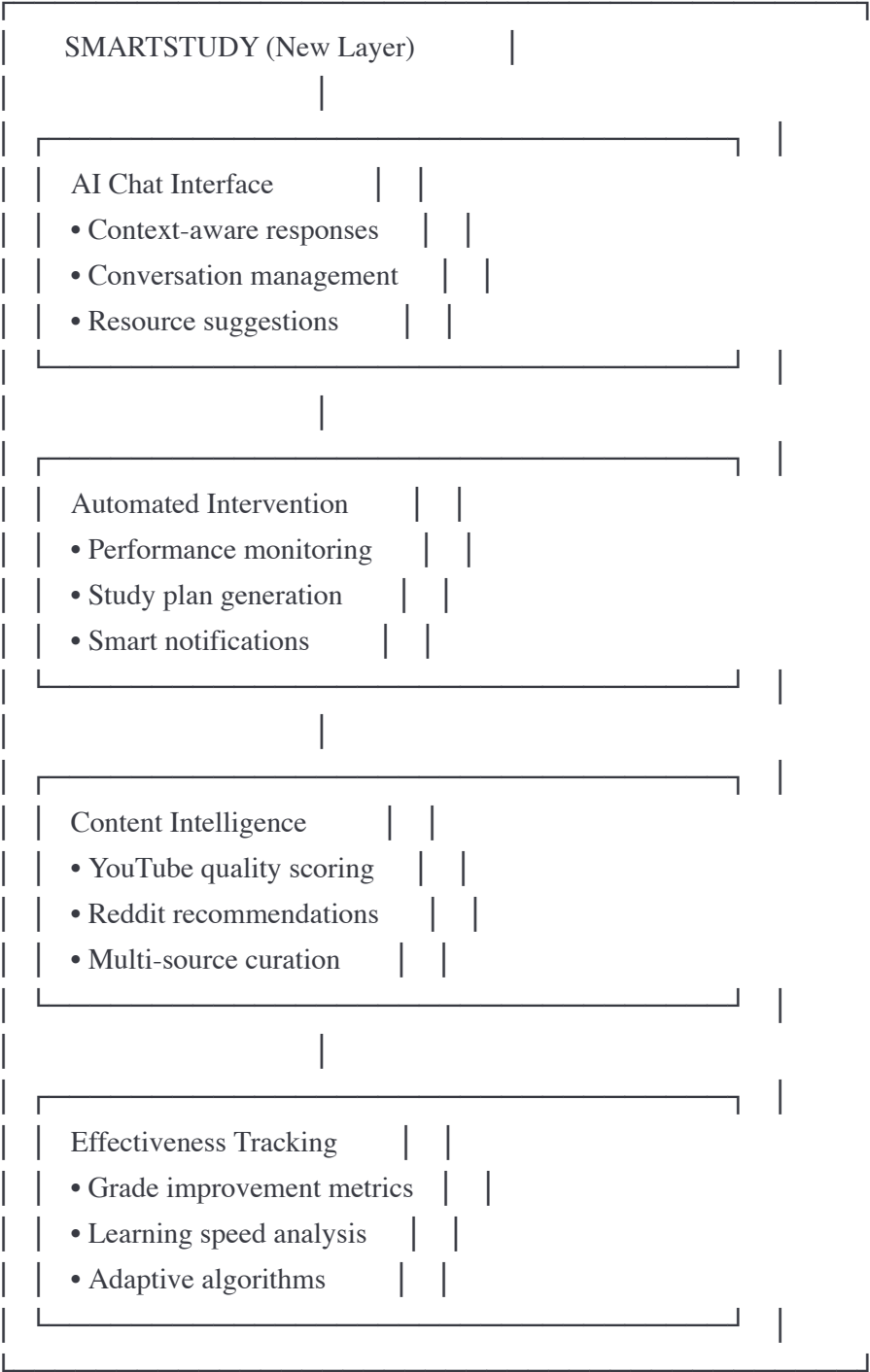
Defense (July 2026)

Presentation: Complete system with measurable impact data

Technical Architecture (High-Level)

Component Overview





Integration Approach

SmartStudy enhances Shadow, doesn't replace it:

- **Chat Access:** New tab in Shadow dashboard
- **Automated Prompts:** Appear contextually in existing interfaces
- **Data Sharing:** SmartStudy reads Shadow's academic data
- **Unified Experience:** Feels like natural extension, not separate app

Cost Analysis

Development Costs

Total: ₦0 (Using existing resources and free APIs)

Operating Costs (Per Semester)

Service	Usage	Cost per Unit	Semester Total
OpenAI GPT-4	~100 study plans + 200 chat sessions	~₦200/plan, ₦50/chat	~₦30,000
YouTube API	Unlimited queries	Free (10K/day limit)	₦0
Reddit API	Unlimited queries	Free (60/min limit)	₦0
Additional hosting	Database storage	Covered by existing	₦0

Total Operating Cost: ~~₦30,000 per semester~~ (\$20 USD)

Extremely cost-effective for a final year project with measurable impact.

Success Metrics (Measurable Outcomes)

Primary Metrics

1. Grade Improvement

- Target: +1.0 letter grade average after using SmartStudy
- Measurement: Compare assessment scores before/after intervention

2. Learning Speed

- Target: 50% faster mastery compared to baseline
- Measurement: Days from struggle detection to proficiency

3. Knowledge Retention

- Target: 75%+ retention after 2 weeks
- Measurement: Follow-up assessment performance

4. Stress Reduction

- Target: 40% decrease in "stressed" mood logs

- **Measurement:** Compare mood patterns before/after intervention

Secondary Metrics

- **Engagement:** 70%+ plan completion rate
- **Satisfaction:** 4.0+ rating out of 5.0
- **Usage:** 3+ logins per week
- **Effectiveness:** 60%+ students meet target grades

Research Value

- Publishable findings on AI-assisted learning effectiveness
- Data on which intervention types work best for different learning styles
- Insights on optimal timing for academic interventions
- Mood correlation with academic performance patterns

Innovation & Academic Merit

Technical Contributions

1. Context-Aware AI Tutoring
 - Novel integration of LLM with real-time academic performance data
 - Personalization beyond generic chatbot responses
2. Automated Content Curation
 - Multi-source quality scoring algorithm
 - Combines engagement metrics, sentiment analysis, and user feedback
3. Effectiveness Measurement Framework
 - Closes the intervention loop with measurable outcomes
 - Adaptive learning based on what actually works
4. Behavioral AI Integration
 - Mood-aware study recommendations
 - Context-sensitive intervention timing

Comparison to Typical Final Year Projects

Typical FYP	Shadow + SmartStudy
Basic CRUD application	Complete academic system + AI intervention

Basic CRUD application	✓ Complete academic system + AI intervention
Simulated/mock data	✓ Real user data over 3-4 months
No AI or basic rules	✓ Advanced LLM integration + behavioral algorithms
No impact measurement	✓ Multiple tracked effectiveness metrics
Generic problem space	✓ Novel solution to identified gap
Theoretical contribution	✓ Practical contribution with research findings

Defense Readiness

Strong points for evaluation committee:

- Real-world problem with clear gap identification
- Novel approach combining multiple AI techniques
- Measurable outcomes with real user data
- Scalable beyond Computer Science to any major
- Publication potential in educational technology
- Demonstrates both technical depth and practical impact

Risk Mitigation Strategies

Risk	Likelihood	Mitigation Strategy
GPT-4 costs exceed budget	Low	Monthly spending cap, caching frequent queries, template fallbacks
LLM generates incorrect advice	Medium	Confidence thresholds, student ratings override, effectiveness tracking flags issues
Low student adoption	Medium	Multiple entry points, proactive prompts, quick wins in first interaction
Cannot complete in timeframe	Low	Modular design - Phase 1 alone is defensible, Phases 2-3 are enhancements
Predictions not accurate	Low	Display confidence levels, allow manual overrides, document limitations clearly
pauarchive incomplete/outdated	High	Multi-source approach - works without pauarchive via YouTube, Reddit, GPT-4
API rate limits	Low	Caching strategies, queue management, degradation gracefully

Why This Enhancement Should Be Approved

1. Addresses Supervisor Request

- Direct response to LLM integration suggestion

- Real AI application with practical value
- Demonstrates advanced technical capability

2. Builds on Strong Foundation

- Shadow already 75% complete and functional
- Adding intelligence layer, not rebuilding
- Realistic 6-month timeline with clear milestones

3. Research Merit

- Novel approach to academic intervention
- Measurable outcomes for effectiveness study
- Potential for publication in educational technology

4. Practical Impact

- Solves real student pain point
- Works for any major, any university
- Sustainable operating costs

5. Defensive Position

- Multiple technical innovations to discuss
- Real user data by defense date
- Clear demonstration of graduate-level work

6. Achievable Scope

- Phase 1 (chat) ships in 1 month
- Progressive enhancement approach
- Clear success criteria at each phase

Integration with Existing Shadow

SmartStudy enhances Shadow's existing features:

Dashboard Integration

Current: Shows CGPA and at-risk courses

Enhanced: Adds "Get Study Plan" button for struggling courses

Course Pages

Current: Displays course grade and tasks

Enhanced: Adds "Ask SmartStudy" for topic-specific help

Task Completion

Current: Logs score and updates CGPA

Enhanced: Prompts intervention if score below target

Mood Logger

Current: Records mood for trends

Enhanced: Adapts study recommendations based on emotional state

All existing functionality remains intact - SmartStudy is an additive layer.

Expected Outcomes by Defense (July 2026)

Quantitative Data

- 20-30 students using SmartStudy
- 300+ chat interactions logged
- 50+ study plans generated
- Grade improvement data from 2 semesters
- Effectiveness metrics across 4 dimensions

Qualitative Evidence

- Student testimonials on impact
- Comparison with baseline academic performance
- Insights on learning style effectiveness
- Recommendations for improvement

Research Findings

- Which intervention types work best
- Optimal timing for academic intervention
- Learning style impact on effectiveness
- Mood correlation with academic outcomes

Demonstration Capability

- Live chat with context awareness
 - Automated intervention triggered by real data
 - Effectiveness dashboard with actual student improvements
 - Resource curation quality comparison
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Comparison with Alternative Approaches

Alternative 1: No LLM Integration (Just Tracking)

Pros: Simpler, cheaper

Cons: Doesn't solve the core problem (students need guidance, not just data)

Alternative 2: Generic ChatGPT Link

Pros: No development needed

Cons: No academic context, no tracking, no integration with Shadow

Alternative 3: Manual Study Plans (Human-Created)

Pros: Potentially higher quality

Cons: Doesn't scale, time-intensive, not adaptive

Proposed Approach: SmartStudy

Pros: Solves core problem, scales infinitely, learns and adapts, measurable impact

Cons: Requires LLM costs (minimal), needs effectiveness validation (built-in)

Conclusion: SmartStudy provides optimal balance of innovation, practicality, and impact.

Next Steps Upon Approval

Immediate Actions (This Week)

1.  Receive supervisor approval on approach
2.  Confirm LLM integration acceptable for FYP guidelines
3.  Set up OpenAI API access and testing environment
4.  Begin Phase 1 implementation (chat interface)

Month 1 (December 2025)

- Build AI chat interface
- Integrate GPT-4 with Shadow's academic context
- Implement conversation management

- Implement conversation management

- Deploy initial version for testing

Month 2 (January 2026)

- Add automated intervention system
- Build content curation algorithms
- Implement study plan generation
- Expand testing to more users

Month 3 (February 2026)

- Add effectiveness tracking
- Implement adaptive learning
- Polish user experience
- Complete core functionality

Months 4-6 (March-June 2026)

- Beta testing with PAU students
- Collect and analyze effectiveness data
- Refine based on real usage
- Prepare defense materials

July 2026

- Present completed system with real impact data
 - Defend technical and research contributions
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Conclusion

SmartStudy represents a significant enhancement to Shadow that transforms it from a diagnostic tool into a treatment system. By integrating Large Language Models as suggested, we can:

1. **Diagnose** why students struggle (not just that they struggle)
2. **Prescribe** personalized learning paths (not generic advice)
3. **Deliver** multi-format content from best sources automatically
4. **Measure** whether interventions actually work (not assume effectiveness)

This addresses a critical gap in academic support systems and positions Shadow as a comprehensive learning companion rather than just a task tracker.

Total Implementation Time: 280-300 hours over 6 months

Operating Cost: ~~₦30,000 per semester~~ (\$20 USD)

Expected Impact: Measurable grade improvements, research findings, publication potential

Request for Approval

I request approval to proceed with SmartStudy implementation as outlined in this proposal. The enhancement aligns with your suggestion to integrate LLMs, addresses a real student need, and maintains a realistic scope for completion within the FYP timeline.

Key Approval Points:

-  LLM integration approach and usage
-  Dual-mode design (chat + automation)
-  Timeline and phased implementation
-  Budget allocation for API costs
-  Beta testing with PAU students

I am prepared to discuss any concerns, adjust the scope as needed, and provide regular progress updates throughout implementation.

Prepared by: Adeleke Paul Aladenusi

Project: Shadow v2.0 - Goal-Driven Academic Achievement System

Institution: Pan-Atlantic University

Expected Defense: July 2026

Contact: [Your Email/Phone]

Appendix: Frequently Anticipated Questions

Q: Why not just use ChatGPT directly?

A: ChatGPT doesn't know the student's academic history, performance data, learning style, or course context. SmartStudy provides this personalized layer that makes responses actionable and trackable.

Q: What if the LLM gives wrong information?

A: Multiple safeguards: (1) Show confidence levels, (2) Student ratings override suggestions, (3) Effectiveness tracking flags poor advice, (4) Template fallbacks for critical scenarios, (5) Human review of patterns.

Q: How does this work for non-CS courses?

A: SmartStudy is discipline-agnostic. Students specify their topics (e.g., "Drug Metabolism" for Pharmacy),

and the LLM adapts. The system learns course patterns over time without requiring pre-loaded syllabi.

Q: Is the operating cost sustainable?

A: At ₦30,000/semester (~\$20), this is extremely affordable. For scale: one student paying ₦300 for better grades makes it cost-effective. Caching strategies further reduce costs.

Q: What if students become dependent on AI?

A: SmartStudy is designed as scaffolding, not replacement. It teaches learning strategies, not just answers. The goal is improved independent learning over time, tracked through effectiveness metrics.

Q: How do you measure if it actually works?

A: Four objective metrics: (1) Grade improvement on next assessment, (2) Faster time to mastery, (3) Knowledge retention after 2 weeks, (4) Stress level reduction. All trackable through Shadow's existing data.

Q: What happens after the FYP defense?

A: SmartStudy can continue operating at minimal cost. The system improves through use (more data = better recommendations). Potential for expansion to other PAU departments or external institutions.

End of Proposal